XRPL Validator — Daily Engineering Report · 2026-04-01

Period: 2026-04-01 00:00 to 24:00 US Eastern (2026-04-01 04:00 to 2026-04-02 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 5 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **perf: flat 256-bucket parallel hash + recursive delete collapse fix** (`ee3356e`, xrpl-validator)
- Replace SHAMap tree with flat sorted array split into 256 buckets - Rayon-parallel compute across 256 buckets: 2.5s total (0.4s merge + 2.1s hash) - spawn_blocking keeps SSE feed smooth during hash computation - Fix SHAMap delete: recursive structural collapse prevents inner→inner→leaf chains - Single-leaf collapse in InnerNode::hash() for correct hash at any tree depth - Gap ledgers use fast path (cache merge …
- **perf: 256-bucket dirty tracking + rippled migration to the reference node** (`3439731`, xrpl-validator) Replace full-array merge with per-bucket binary search updates (0.65s → 0.03s). Only recompute dirty bucket hashes via rayon (skip ~30% clean buckets). Update RPC endpoints from the validator host localhost to the reference node ([host]).
- **feat: network resilience — RPC failover + endpoint migration** (`c90dde8`, xrpl-validator)
- rippled_client.rs: update primary endpoint to the reference node ([host]) - bulk_sync.rs: replace hardcoded RPC_URL with rpc_failover() that cycles through local → xrplcluster → s1.ripple.com - All source files: migrate hardcoded [host] → [host] - rippled now runs on the reference node (NVMe, systemd, 3.1.2) - Validator on the validator host has full CPU with no rippled contention
- **feat: persist leaf cache to disk for instant restarts** (`160db5e`, xrpl-validator)
- FlatHasher::save(): atomic write (tmp + rename) of 1.2GB binary cache - FlatHasher::load(): reads cache + recomputes bucket hashes in 2.5s - Auto-save every 500 matches in background thread - SIGTERM handler saves cache before exit - Cache path set next to RocksDB (leaf_cache.bin) - Restart time: 30s → 2.5s (12x faster)

Docs and operations

- **docs: update README to reflect current validator status** (`81295d1`, xrpl-validator)

Independence program

No independence record for UTC day 2026-04-01; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-01 04:47	—	xrpl-validator	`ee3356e`	perf: flat 256-bucket parallel hash + recursive delete collapse fix
04-01 22:33	—	xrpl-validator	`81295d1`	docs: update README to reflect current validator status
04-02 00:37	—	xrpl-validator	`3439731`	perf: 256-bucket dirty tracking + rippled migration to the reference node
04-02 00:59	—	xrpl-validator	`c90dde8`	feat: network resilience — RPC failover + endpoint migration
04-02 01:40	—	xrpl-validator	`160db5e`	feat: persist leaf cache to disk for instant restarts

Times in UTC. A commit appears on the day it was committed to main or merged into it.